

More than two pins

Real nets have many pins

The idea

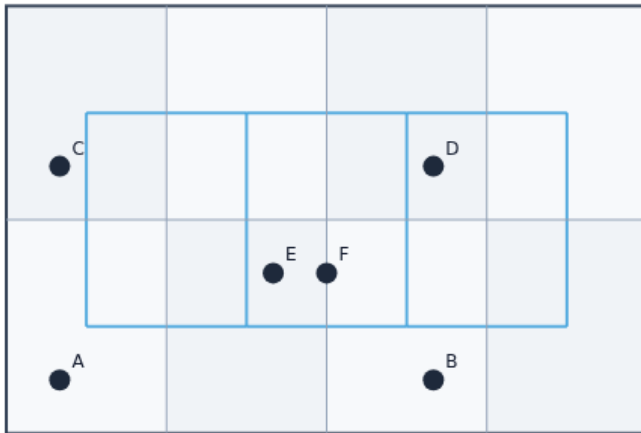
- Collect terminal GCells for the net
- Center is the integer average of min and max column and row
- For each pin, add the edges along l_route center to pin with HV unless you choose VH
- Union all legs, edges shared by two legs count once in usage when you deposit

Four-pin net

MULTI-PIN TREE

Four-pin net

Step 1 / 5 · net4 · module02-07-multipin-tree



Explanation

Net [A,B,C,D] needs a tree, not a single two-pin path.

- Star demo
- 4 legs

Net index 4

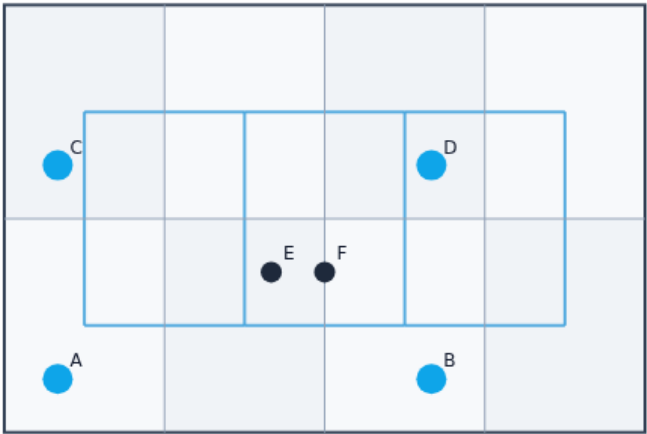
Four-pin net

Bbox center

MULTI-PIN TREE

Bbox center

Step 2 / 5 · center · module02-07-multipin-tree



Explanation

Star hub at mean GCell of pins (clamped).

- Steiner lite
- Toy only

Hub GCell

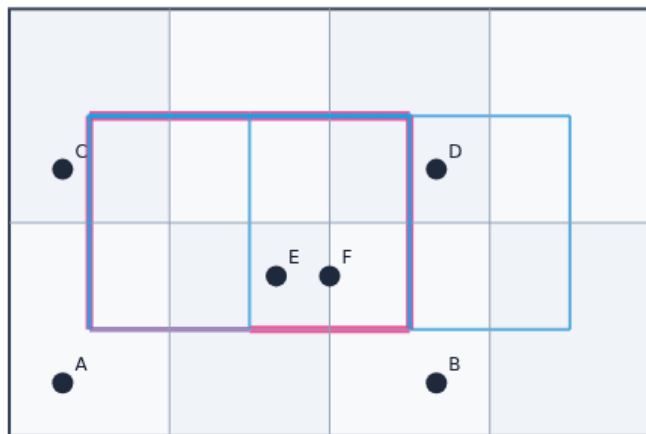
Bbox center

Four L legs

MULTI-PIN TREE

Four L legs

Step 3 / 5 · legs · module02-07-multipin-tree



Explanation

Each pin gets an L-route from the hub in the full sequential pass.

- 4 paths
- Edge sharing

Highlight net 4

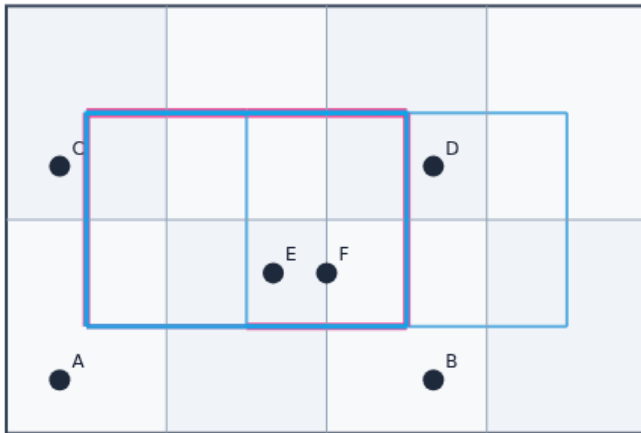
Four L legs

Shared edges

MULTI-PIN TREE

Shared edges

Step 4 / 5 · usage · module02-07-multipin-tree



Explanation

Multi-pin trees reuse edges—overflow adds up.

- Sum usage
- Cap=2

Watch corridors

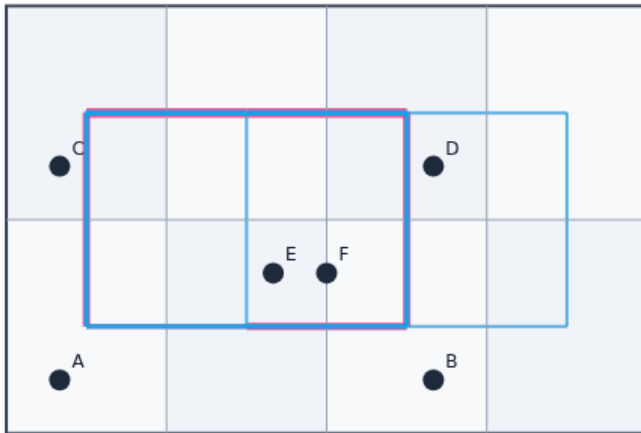
Shared edges

Two-pin nets

MULTI-PIN TREE

Two-pin nets

Step 5 / 5 · two · module02-07-multipin-tree



Explanation

Short nets E-F stay single L paths.

- Mix topologies
- Six nets total

E-F short

Two-pin nets

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Multi-pin Steiner tree

INTERACTIVE TOOL

Multi-pin Steiner tree

Four-pin net A–B–C–D routes as a star from the bbox center GCell.

Your job: net4 is [A,B,C,D]. Route L-HV; the star uses four legs from center.

Reset workspace

Challenges0 / 8 cleared

Pick one [Intro] Net4 listed

Pick a challenge and click **Start**. Move cells, route, then **Check**. Reveal golden is study-only.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A

B

C

D

E

F

←

→

↑

↓

Reset to starter

Reveal golden (study)

Route L-HV

Route L-VH

Route maze

Rip-up reroute

Graph

Reference / metrics

selected: A @ GCell (0,0)
routes: (none – click Route)
overflow total: 0.00 · max: 0.00 · count: 0
edge capacity: 2
HPWL: 34.0
view: your placement

Browser lab starter

learn_global_routing — 07 multipin tree

Implement track

- Implement ``multipin_star`` and ``multipin_star_edges``
- Route net index four on `tiny_gr` and print hub GCell plus edge count

Pitfalls

- Routing pairwise between every pin pair, that explodes edges
- Picking chip center instead of bbox center GCell
- Double-depositing the same edge when summing legs

Your turn

- Complete multipin routing
- Next: quantify overflow on those shared edges